Pre-analysis Plan Many Daughters Project

Hypothesis 1: Having daughters leads to stronger support for gender equality (as measured by attitudes)

Variables

To test this hypothesis, I will use variable plh0303, which ask respondents whether men should be involved in housework, as the dependent variable. This is a four-point ordinal variable ranging from 'agree completely' to 'disagree completely. The variable would be coded so higher values reflect higher support (1= disagree completely, 4= agree completely).

As with the rest of the hypothesis, the main independent variable would be whether a person has a daughter. To code this, I will rely on the variables kidsex01- kidsex019. Which would take a value of 1 if any of those variables is equal to 'female', and 0 otherwise.

Exclusion criteria

I would include all observations for which the data is available. The only criterion of exclusion is having children. As long as the respondent has one child, they would be included in the sample. By selecting on the having children, the benchmark becomes having children *vs* having a daughter. This will help better isolate the effect of having daughters, as opposed to combining the potential effect of having children and having daughters. This approach also helps establish a randomly assigned treatment since having daughters is random given the decision of having children.

Analysis plan

The data will be analyzed with generalized linear model, specifically with ordinal logit. This allows the model to better reflect the structure of the dependent variable. The only predictor in the regression would be having a daughter. Since the treatment is randomly assigned, I do not foresee any potential for confounding. While other variables may affect values of the dependent variable, they will not affect the values of the independent variable (given the selection on having children). This precludes the need for additional controls or the incorporation of any clustering structure in the data.

Interpretation of results

I will rely on standard null-hypothesis framework to interpret the results. I will treat a *p*-value of less than 0.5 for the independent variable as evidence that the hypothesis cannot be rejected.

Hypothesis 2: Having daughters leads to more left-leaning political preferences.

Variables

The dependent variable for this model will be based on plh0012_h, which asks people about their party affiliation ('Which party do you lean toward?'). It will be coded as 1 if the respondent reports any supporting the SPD, Die Lenke, or Bündnis 90. Then, I will multiply that resulting variable with the variable plh0013_h, which asks respondents how much they support the party they mentioned. I will recode plh0013 so that higher values imply higher support. The resulting variable will be an ordinal one that ranges from 0 to 5.

Exclusion criteria

Same as for Hypothesis 1.

Analysis plan

Same as hypothesis 1, but the p-value of interest will be that of the new predictor (party affiliation).

Interpretation of results

Same as for Hypothesis 1.

Hypothesis 3: Having daughters leads to more gender equality in the household as measured by behaviors.

Variables

I will rely on question plh0303 as the dependent variable for the test of this hypothesis. This question asks respondents "How many hours per normal work day do you spend on housework (washing, cooking, cleaning)".

For the independent variables, I will use two predictors. The first one will be having a daughter as measured in hypothesis 2. The second one will be household size (hhgr).

Exclusion criteria

Same as for Hypothesis 1.

Analysis plan

Same as Hypothesis 1, but with an added interaction term between the two predictors. This addition of this second variable aims to capture the fact that the time someone spends on housework is partly determined by whether they are the only person responsible for it. So, even if someone has a daughter and has a strong belief in gender equality, they might not need to spend as much time on housework if there is someone else who shares that responsibility. In other words, household size could moderate the effect of having a daughter on the deponent variable.

Interpretation of results

If the effect of having a daughter is statistically significantly different from zero, then I would consider that as evidence that the null hypothesis cannot be rejected.

Hypothesis 4: Having daughters leads to a larger increase in the support of gender equality for men than for women as measured by attitudes.

Variables

Same as hypothesis 1 but with an added predictor for the gender of the respondent pla0009, recoded so 1=female, 0 otherwise.

Exclusion criteria

Same as hypothesis 1

Analysis plan

Same as hypothesis 1 but with an added interaction term between having a daughter and gender.

Interpretation of results

I will compare the marginal effect of having a daughter between the two groups (men and women), and see if the former is statistically significant bigger than zero. If so, I will conclude that null hypothesis cannot be rejected.